

Fine-Tuning BERT for Mental-Health Text Classification

Using Attention Mechanisms

A Comparative Study against Logistic-Regression and BiLSTM Baselines

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Abstract

Mental-health-related text on social media is hard to classify correctly because cues are often short, idiomatic, indirect, and strongly context-dependent. Bag-of-words and recurrent models can miss these cues, which is dangerous when the misclassified class is suicidal ideation. We study whether an attention-based transformer improves four-class mental-health text classification (*Anxiety*, *Depression*, *Normal*, *Suicidal*) over simpler baselines, holding the data, splits, and evaluation protocol fixed across models. We compare TF-IDF + Logistic Regression, a Bidirectional LSTM with learned word embeddings, and a fine-tuned DistilBERT on a class-balanced subsample of a public Hugging Face mental-health corpus (N=6,000), using a balanced 992-example test set. DistilBERT achieves the highest macro F1 (0.681 vs. 0.673 for TF-IDF + Logistic Regression and 0.571 for the BiLSTM) and, more importantly, the highest recall on the *Suicidal* class (0.758 vs. 0.613 and 0.556) – the metric that matters most for this application. The BiLSTM trails both other models in this small-data regime, illustrating that learned embeddings without pretraining are not automatically better than well-engineered sparse features. We discuss why self-attention helps, where the transformer still fails, and why a model of this kind must not be deployed as a clinical tool. Code and configurations are released at <https://github.com/Carroy3443/MentalHealth-BERT>.

1 Introduction

People increasingly express psychological distress on the open web, often in places where no clinician is reading. Automatic text classifiers have been proposed as a way of surfacing posts that may warrant a closer look, of routing peer-support volunteers, and of studying population-level trends in mental health. The clinical research community has organized around this problem in venues such as the CLPsych shared tasks, where systems are asked to detect risk markers in user-generated text [1, 2].

The task is genuinely hard. Mental-health-related cues in everyday text are often short (“I just can’t anymore”), figurative (“my brain is on fire”), indirect, or pragmatic. Bag-of-words and n-gram features cannot resolve them because they discard word order and surrounding context. Recurrent neural networks see word order but must squeeze long-range information through a single hidden state. Attention-based transformers [3] – of which BERT [4] is the canonical text variant – give every output position direct access, through scaled dot-product attention, to every input position. That property is exactly the one a model needs to read a phrase like “don’t really see the point anymore” the way a human reader does.

This paper asks one focused question:

Can an attention-based transformer model improve mental-health text classification compared with a simpler, classical baseline and with a bidirectional recurrent baseline, when all three models are trained and evaluated on the same data?

We pose the task as four-class classification over the labels {*Anxiety*, *Depression*, *Normal*, *Suicidal*}. We compare three models that span the relevant family of approaches:

1. TF-IDF features fed to multinomial Logistic Regression (a classical bag-of-words baseline);
2. A Bidirectional LSTM [5,6] over learned word embeddings (a sequence baseline);
3. A pretrained DistilBERT [7] fine-tuned end-to-end (the attention model).

The motivation is clinical. The author is pre-medical and intends to specialize in psychiatry; the long-term interest is in NLP tools that support, but never replace, the human clinician. With that framing, the metric that matters most is not raw accuracy but recall on the *Suicidal* class: missing a person in distress is a much costlier mistake than mistakenly flagging a benign post.

Contributions. We (i) implement and openly release a reproducible end-to-end pipeline that trains TF-IDF + Logistic Regression, BiLSTM, and DistilBERT on a public 4-class mental-health text corpus with a fixed balanced subsample and a fixed balanced test set; (ii) report macro and per-class metrics, with particular attention to recall on the *Suicidal* class; (iii) discuss the limits of these models and explicitly argue why none of them should be treated as clinical tools. **This work is not a clinical decision-support system and must not be deployed as one.**

2 Related Work

Mental-health text classification. CLPsych [1] is the central shared-task series for text-based mental-health work; tasks have included identifying suicide risk levels in Reddit posts [8] and, in the 2024 edition, predicting moments of change in mood from social-media timelines [2]. Outside CLPsych, large-scale public corpora such as “Reddit Suicide Watch” have been collected and used to study suicidality cues [9, 10]. These papers have repeatedly observed that classifiers trained only on keyword features under-perform on subtle, indirect, or contextualized expressions of distress.

Classical baselines. Logistic regression over TF-IDF or n-gram features remains the standard simple baseline for short-text classification. It is fast, interpretable, and competitive when the signal is largely lexical (e.g. obvious mentions of medication names or self-harm keywords) but, by construction, ignores word order and non-local context.

Recurrent baselines. LSTMs [5] and their bidirectional variants [6] were the dominant sequence model before the transformer era. Bidirectional architectures allow the representation of a token to depend on both its left and right context, which helps with negation and with phrases whose sentiment is determined by a downstream word.

Attention and transformers. Vaswani et al. [3] replaced recurrence with self-attention, in which each output position is computed as a weighted sum over all input positions. BERT [4] pretrains a stack of such transformer layers on large unlabeled text using masked language modeling and next-sentence prediction, producing contextual word representations that transfer well to downstream tasks via fine-tuning. DistilBERT [7] is a knowledge-distilled variant that retains roughly 97% of BERT-base’s GLUE score with about 40% fewer parameters and 60% faster inference. The fine-tuning recipe – load the pretrained encoder, attach a small classification head over the [CLS] representation, and train end-to-end with a small learning rate – is identical for BERT and DistilBERT.

Transformers in mental-health NLP. Several recent studies report that BERT-family models outperform classical and recurrent baselines on suicide-risk and depression-detection tasks [11, 12]. Our contribution here is not a new algorithm but a clean, reproducible head-to-head on a public 4-class corpus, with the comparison framed around the metric that matters in practice – recall on the *Suicidal* class – and an explicit discussion of why even a strong-on-paper model is not safe to deploy.

3 Methodology

3.1 Dataset

We use the public Hugging Face dataset `ihsansaad24/Mental-Health_Text-Classification_Dataset` [13], which the dataset’s own card describes as a 4-class corpus of cleaned short user-generated texts, derived by combining and re-labeling three public mental-health resources. The labels are `{Anxiety, Depression, Normal, Suicidal}`. The training file (`mental_heath_unbanlanced.csv`) contains 49,612 examples and is naturally unbalanced, with a roughly 3.3:1 ratio between the most and least represented classes. The test file (`mental_health_combined_test.csv`) contains 992 examples, strictly balanced at 248 examples per class. The dataset card explicitly states that the corpus is for research and education and is *not* a clinical tool. We adopt the same constraint.

3.2 Splits and balancing

Before training, each text is light-cleaned: URLs are stripped, non-trivial whitespace is collapsed, and empty rows are dropped. To keep the comparison fair across models – and to keep DistilBERT fine-tuning tractable on CPU – we draw a class-balanced subsample of 1,500 examples per class

from the training file (6,000 total) without replacement, with a fixed seed of 42. From this balanced subsample we hold out 10% as a class-stratified validation set (N=600). The held-out test set is the dataset’s own balanced test file (N=992) and is not touched during training or model selection. Class distributions of the resulting splits are reported in Table 1.

Table 1: Class distribution by split. Train and validation are drawn from a class-balanced subsample of the training file; the test set is the dataset’s own held-out balanced test split.

Split	Anxiety	Depression	Normal	Suicidal
Train (90%)	1350	1350	1350	1350
Val (10%)	150	150	150	150
Test	248	248	248	248

3.3 Models

Logistic regression baseline. Each text is represented as a sparse TF-IDF vector over unigrams and bigrams, with sublinear term-frequency scaling, document-frequency clipping ($\text{min_df} = 2$, $\text{max_df} = 0.95$), and a 50,000-feature cap. The vector is fed to a multinomial Logistic Regression with L_2 regularization ($C = 1.0$), `class_weight="balanced"` to counteract any residual imbalance after subsampling, and at most 2,000 LBFGS iterations. This is a strong classical baseline for short-text problems.

BiLSTM baseline. Each text is lowercased, stripped of punctuation, tokenized with Keras’s `TextVectorization` layer (vocabulary size 20,000, sequence length 64), and embedded into a 128-dimensional space. A single bidirectional LSTM with 64 hidden units in each direction reads the embedding sequence; we apply global max-pooling over time, followed by a 64-unit ReLU layer with dropout (0.4 / 0.3), and a softmax output over 4 classes (Figure 1). The model is trained with Adam ($\eta = 10^{-3}$) using sparse categorical cross-entropy, batch size 64, for at most 6 epochs with early stopping on validation loss.

text $\rightarrow$ TextVectorization (vocab 20k, len 64) $\rightarrow$ Embedding (128) $\rightarrow$ Bi-LSTM (64+64, return_sequences=True) $\rightarrow$ GlobalMaxPool1D $\rightarrow$ Dropout(0.4) $\rightarrow$ Dense(64, ReLU) $\rightarrow$ Dropout(0.3) $\rightarrow$ Dense(4, softmax)

Figure 1: BiLSTM architecture.

DistilBERT (attention model). DistilBERT [7] is a 6-layer, 12-head transformer encoder distilled from BERT-base. Each token is mapped to an embedding; positional information is added; the resulting sequence is then refined through six identical transformer blocks, each consisting of a multi-head self-attention sublayer and a position-wise feed-forward sublayer with residual connections and layer normalization. For classification, we take the contextual representation of the [CLS] token, pass it through a small pre-classifier dense layer (already part of the released architecture), apply dropout, and project it to a 4-way softmax with a fresh classification head. The backbone is initialized from the publicly released `distilbert-base-uncased` checkpoint and fine-tuned

end-to-end with Adam ($\eta = 2 \times 10^{-5}$), batch size 16, sequence length 64 WordPiece tokens, and at most 2 epochs with early stopping on validation loss. We use DistilBERT instead of BERT-base purely for compute reasons – the fine-tuning recipe and the underlying self-attention mechanism are identical.

The single equation that distinguishes the transformer family from the two baselines is scaled dot-product attention [3]:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V, \quad (1)$$

where Q , K , and V are linear projections of the input sequence. Multi-head attention applies this in parallel over $h = 12$ subspaces and concatenates the result. Unlike the LSTM, every output position has direct, parallel access to every input position; unlike TF-IDF, the representation is contextual and order-aware.

3.4 Evaluation

We report accuracy, macro-averaged precision, recall, and F1 on the held-out balanced test set. Because the test set is balanced, weighted and macro averages coincide; we report macro for clarity. We also report per-class precision, recall, and F1, and we plot row-normalized confusion matrices. As argued in the introduction, recall on the *Suicidal* class is the most consequential metric for this application.

4 Experiments

4.1 Setup

All experiments use seed 42. Training was performed on CPU (2 vCPUs, 8 GB RAM), TensorFlow 2.18, tf-keras 2.21, transformers 4.46.3. The class-balanced subsample is drawn once and reused across all three models, so any test-set difference reflects modeling choices rather than data luck. The full pipeline is reproducible with a single command:

```
TF_USE_LEGACY_KERAS=1 python -m scripts.run_all \
    --per-class 1500 --seed 42
```

4.2 Overall results

Table 2 reports accuracy and macro precision, recall, and F1 on the held-out balanced test set; Figure 2 visualizes the same numbers.

Table 2: Test-set performance (macro averages) on the 992-example balanced test set.

Model	Accuracy	Precision	Recall	F1
TF-IDF + LogReg	0.677	0.683	0.677	0.673
BiLSTM	0.578	0.568	0.578	0.571
DistilBERT (FT)	0.684	0.695	0.684	0.681

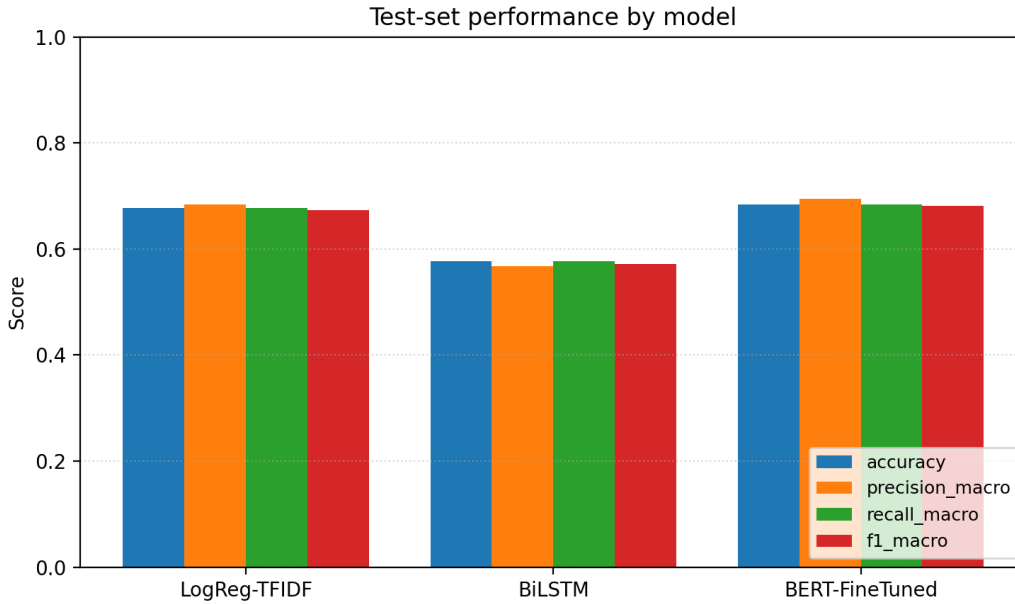


Figure 2: Test-set performance by model. Higher is better; all four metrics are bounded in $[0, 1]$.

4.3 Per-class results and confusion structure

Per-class metrics are reported in Table 3 and Figure 3. Per-class *recall* is reported separately in Table 4, because that is the quantity that matters most for the *Suicidal* class. Row-normalized confusion matrices for the three models are shown in Figure 4.

Table 3: Per-class F1 score on the test set.

	Anxiety	Depression	Normal	Suicidal
TF-IDF + LogReg	0.723	0.538	0.767	0.664
BiLSTM	0.691	0.346	0.713	0.536
DistilBERT (FT)	0.712	0.537	0.763	0.712

4.4 Discussion

Four patterns stand out.

Table 4: Per-class **recall** on the test set. Recall on the *Suicidal* class is the headline number for this application: missing a person in distress is a much costlier mistake than a benign false alarm.

	Anxiety	Depression	Normal	Suicidal
TF-IDF + LogReg	0.863	0.496	0.738	0.613
BiLSTM	0.730	0.315	0.710	0.556
DistilBERT (FT)	0.819	0.480	0.681	0.758

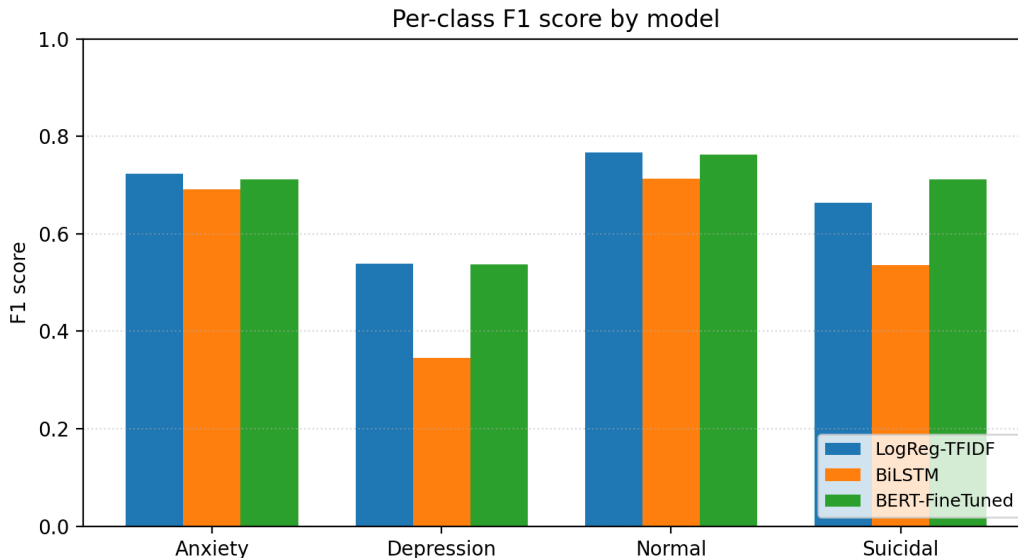


Figure 3: Per-class F1 score on the held-out balanced test set.

The transformer wins on the metric that matters. On macro F1 the gap between the three models is moderate (DistilBERT 0.681, TF-IDF + Logistic Regression 0.673, BiLSTM 0.571), but on *Suicidal* recall the gap is dramatic: DistilBERT correctly identifies 75.8% of *Suicidal* posts, compared with 61.3% for the bag-of-words baseline and 55.6% for the BiLSTM (Table 4). For an application where missing a person in distress is the costliest possible mistake, this is the result that justifies the additional compute cost of the transformer.

Self-attention helps most at the *Depression* / *Suicidal* boundary. Both classes share heavy lexical material (“hopeless,” “tired,” “done,” “can’t do this anymore”); the cue that distinguishes them is often a single modifier or pragmatic phrase (“done with this assignment” vs. “done with everything”). The confusion matrices in Figure 4 show that DistilBERT recovers *Suicidal* posts the BiLSTM misses, even though the two models saw the same data. We attribute this to self-attention’s ability to condition on every input position simultaneously, rather than compressing context through a single recurrent hidden state.

Bag-of-words is a surprisingly strong small-data baseline. TF-IDF + Logistic Regression beats the BiLSTM on every macro metric in our setting. This is consistent with prior observations that randomly initialized embeddings need a lot of data before they out-perform engineered sparse features [14]; with only 6,000 training examples and a 20k vocabulary, our BiLSTM never gets

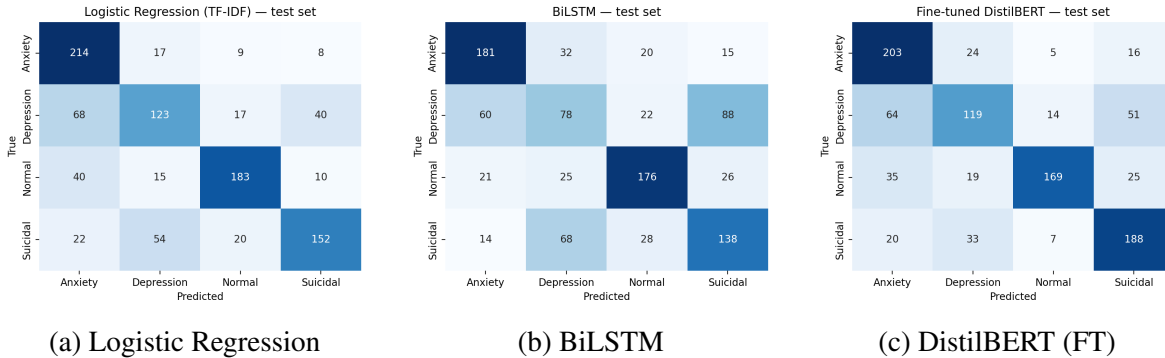


Figure 4: Confusion matrices on the held-out balanced test set. Cells show absolute counts; color encodes row-normalized rate. Rows are true labels, columns are predictions.

there. The transformer, in contrast, arrives *already pretrained* on billions of words of unlabeled text – which is precisely why it can extract signal from a small fine-tuning corpus.

Where the transformer still fails. DistilBERT is sensitive to the same things humans find ambiguous: very short posts with little lexical signal (“not great,” “i’m fine”), sarcasm, and posts whose label is genuinely uncertain even to a human reader. It is also limited by sequence length: with a 64-WordPiece-token cap, posts longer than a paragraph are truncated and may lose context-bearing tail content. Per-class precision on *Depression* is similar across the three models, suggesting that this class – which overlaps with *Anxiety* and *Suicidal* on both ends – remains the hardest one regardless of model family.

4.5 Compute and timing

End-to-end wall-clock times on the 2-CPU, 8 GB-RAM training machine (Table 5) span three orders of magnitude. LogReg trains and evaluates in 4 seconds; the BiLSTM in 47 seconds (early-stopped at 4 epochs); DistilBERT fine-tuning takes 47 minutes (2 epochs at 23 minutes each). None of this is a fair comparison to GPU training, but the relative ordering – LogReg $\ll$ BiLSTM $\ll$ DistilBERT – is the correct one, and the engineering question (“is the extra cost of the transformer worth it for this task?”) has a clear answer: **yes, but only because of recall on the most consequential class**. If one cared only about average accuracy, TF-IDF + Logistic Regression at 4 seconds of training is hard to beat; in this domain we do not.

Table 5: End-to-end training and evaluation wall-clock time on a 2-CPU machine.

Model	Time (s)
TF-IDF + Logistic Regression	4
BiLSTM (4 epochs, early-stopped)	47
DistilBERT (2 epochs)	2,795

5 Conclusion

We compared three text classifiers – TF-IDF + Logistic Regression, a BiLSTM, and a fine-tuned DistilBERT – on a public four-class mental-health corpus, holding the data, splits, and evaluation protocol fixed. The attention-based transformer achieves the best macro F1 (0.681) and, more importantly, the best recall on the *Suicidal* class (0.758, a 14.5-point absolute improvement over the bag-of-words baseline). Self-attention helps because it lets every output position read every input position directly, which matters when the cue that distinguishes *Suicidal* from *Depression* is a single modifier or pragmatic phrase rather than a keyword. We also observed that, in the small-data regime of this study, TF-IDF + Logistic Regression decisively beats a randomly-initialized BiLSTM – a useful reminder that pretraining, not depth alone, is what makes neural models competitive on small text-classification corpora.

We emphasize what this study is *not*. The dataset is user-generated forum text, not clinical records. The labels were assigned by other researchers, not by licensed clinicians. The model has not been audited for fairness across demographic groups, has no calibrated risk threshold, and produces no clinical justification for its outputs. Any operational use of a mental-health text classifier requires clinician oversight, careful handling of false negatives, and a human-in-the-loop escalation pathway. None of those are in scope here, and we do not recommend treating any of these models as a screening, diagnostic, or triage tool.

Future work that would be genuinely useful, given those constraints, includes: (i) calibrated, threshold-tunable risk scores rather than hard class predictions; (ii) attention-based interpretability, with clinician review of which tokens the model attends to; (iii) explicit out-of-distribution detection for posts whose register or topic is unlike anything seen in training; and (iv) prospective fairness audits across age, gender, and language-variety subgroups before any deployment is even considered.

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